

Week 2 : Product and Category Management System

Scenario:

The E-commerce company wants to maintain information about product and their categories. Each product belongs to a specific category and the system should manage product details such as name, price, and stock availability.

1.Design Product and Category Tables

Product Table

```
SQL> CREATE TABLE PRODUCT(Product_ID VARCHAR2(10) PRIMARY KEY,Product_Name VARCHAR2(15) NOT NULL,Category_ID VARCHAR2(10),Price NUMBER(10,2),CustomizationAvailable CHAR(1) DEFAULT 'Y',ProductImage VARCHAR2(255));
```

Table created.

Category Table

```
SQL> CREATE TABLE Category(Category_ID VARCHAR2(10),Category_Name VARCHAR2(20),Description VARCHAR2(50),CustomizationAvailable CHAR(1) DEFAULT 'Y',CategoryImage VARCHAR2(300));
```

Table created.

Two tables for category and product have been created.

Now we have to find the relationship between the tables and apply constraints to the tables.

2.Define Relationships and Constraints

Relationships:

1. The **Category** table stores different ceramic product categories such as Kitchenware, Garden_Outdoor, and Home_Decor.
2. The **Product** table stores individual ceramic products belonging to these categories.
 - **Primary Key:** A Primary Key is a column that uniquely identifies each record in a table.
 - **Category_ID** is the **Primary Key** in Category Table
 - **Foreign Key :** The Key which links two tables.
 - **Category_ID** is **Foreign Key** in Product Table
3. Therefore, the relationship is **One-to-Many (1:N)** — one category can have many products.

Constraints

- ✓ **PRIMARY KEY** – Uniquely identifies each record.
- ✓ **FOREIGN KEY** – Connects Product with Category.
- ✓ **NOT NULL** – Ensures a value must be entered.
- ✓ **CHECK** – Allows only valid values, such as Y or N.
- ✓ **UNIQUE** – Prevents duplicate values in a column.

3. Define Relationship and Create Constraints for Product and Category.

```
SQL> ALTER TABLE Category
  2  ADD CONSTRAINT PK_Category PRIMARY KEY (Category_ID);

Table altered.
```

✓ **PRIMARY KEY** – Uniquely identifies each record

UNIQUE – Prevents duplicate values in a column

```
SQL> ALTER TABLE CATEGORY
  2  ADD CONSTRAINT UQ_Category_Name UNIQUE (Category_Name);

Table altered.
```

4. Insert Values:

For Category Table

```
SQL> INSERT INTO Category
  2  (Category_ID, Category_Name, Description, CustomizationAvailable, CategoryImage)
  3  VALUES
  4  ('00001',
  5  'Kitchenware',
  6  'Durable ceramic products for everyday use',
  7  'Y',
  8  'Elegant ceramic kitchenware collection');

1 row created.
```

```
SQL> INSERT INTO Category
  2  (Category_ID, Category_Name, Description, CustomizationAvailable, CategoryImage)
  3  VALUES
  4  ('00002',
  5  'Garden_Outdoor ',
  6  'Ceramic products for gardens and outdoor spaces',
  7  'Y',
  8  'Elegant ceramic garden collection featuring colorful flower pots, decorative planters, garden ornaments, and beautiful outdoor ceramic pieces surrounded by green plants and flowers');

1 row created.
```

```
SQL> INSERT INTO Category
  2  (Category_ID, Category_Name, Description, CustomizationAvailable, CategoryImage)
  3  VALUES
  4  ('00003',
  5  'Home_Decor ',
  6  'Elegant ceramic products for every home ',
  7  'Y',
  8  'Decorative ceramic vases, figurines, wall pieces, and artistic ornaments displayed in a modern living room');

1 row created.
```

Result

```
Connected.
SQL> SELECT * FROM category;

CATEGORY_I CATEGORY_NAME
-----
DESCRIPTION                                C
-----
CATEGORYIMAGE
-----
00003      Home_Decor
Elegant ceramic products for every home      Y
Decorative ceramic vases, figurines, wall pieces, and artistic ornaments display
ed in a modern living room

00001      Kitchenware
Ceramic products for kitchen use              Y

CATEGORY_I CATEGORY_NAME
-----
DESCRIPTION                                C
-----
CATEGORYIMAGE
-----
Ceramic plates, cups, bowls, and kitchen items

00002      Garden_Outdoor
Ceramic products for gardens                  Y
Ceramic pots, planters, and outdoor garden decor
```

For Product Table

```
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
  3 VALUES
  4 ('P00001', 'Dinner_Plate', '00001', 299.00, 'Y',
  5  'Elegant white ceramic dinner plate with a smooth finish displayed on a dining table');

1 row created.

SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
  3 VALUES
  4 ('P00004', 'Ceramic_Flower_Pot', '00002', 399.00, 'Y',
  5  'Colorful ceramic flower pot filled with a green plant and displayed in a beautiful garden');
('P00004', 'Ceramic_Flower_Pot', '00002', 399.00, 'Y',
*)
ERROR at line 4:
ORA-12899: value too large for column "ECOMMERCE"."PRODUCT"."PRODUCT_NAME"
(actual: 18, maximum: 15)

SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
  3 VALUES
  4 ('P00004', 'Flower_Pot', '00002', 399.00, 'Y',
  5  'Colorful ceramic flower pot filled with a green plant and displayed in a beautiful garden');

1 row created.

SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
  3 VALUES
  4 ('P00002', 'Coffee Mug', '00001', 199.00, 'Y',
  5  'Beautiful ceramic coffee mug with a stylish design placed on a wooden kitchen table');

1 row created.
```

```

1 row created.

SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
3 VALUES
4 ('P00003', 'Garden Planter', '00002', 599.00, 'Y',
5 'Decorative ceramic garden planter surrounded by green plants and colorful flowers outdoors');

1 row created.

SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
3 VALUES
4 ('P00005', 'Soup Bowl', '00003', 249.00, 'Y',
5 'Elegant ceramic soup bowl with a smooth finish displayed beautifully on a dining table');

1 row created.

SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Category_ID, Price, CustomizationAvailable, ProductImage)
3 VALUES
4 ('P00006', 'Dessert Plate', '00003', 229.00, 'N',
5 'Stylish ceramic dessert plate with a simple modern design for serving sweet dishes');

1 row created.

```

Result

```

SQL> SELECT * FROM Product;

PRODUCT_ID PRODUCT_NAME      CATEGORY_I      PRICE C
-----
PRODUCTIMAGE
-----
P00001      Dinner_Plate      00001           299 Y
Elegant white ceramic dinner plate with a smooth finish displayed on a dining ta
ble

P00004      Flower_Pot        00002           399 Y
Colorful ceramic flower pot filled with a green plant and displayed in a beautif
ul garden

P00002      Coffee Mug        00001           199 Y

PRODUCT_ID PRODUCT_NAME      CATEGORY_I      PRICE C
-----
PRODUCTIMAGE
-----
Beautiful ceramic coffee mug with a stylish design placed on a wooden kitchen ta
ble

P00003      Garden Planter    00002           599 Y
Decorative ceramic garden planter surrounded by green plants and colorful flower
s outdoors

P00005      Soup Bowl         00003           249 Y
Elegant ceramic soup bowl with a smooth finish displayed beautifully on a dining

PRODUCT_ID PRODUCT_NAME      CATEGORY_I      PRICE C
-----
PRODUCTIMAGE
-----
table

P00006      Dessert Plate     00003           229 N
Stylish ceramic dessert plate with a simple modern design for serving sweet dish
es

```

As for we have created two table product and category and inserted values and constrains for both tables.